

Compact Objects as Almost-Perfect All:All Entanglement Graphs: From fast scrambling and weak-field gravity to a testable gap-kilonova prediction

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`python scripts/generate_figures.py` reproduces every figure; 2194 tests

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Abstract

We study a phenomenological model in which spacetime connectivity is an entanglement graph and a black-hole interior is an *almost-perfect* all:all (complete) subgraph. Interior edges cost no exterior space; each of k exterior legs costs $4 \ln 2 \approx 2.77$ Planck patches of horizon area, so $A(k) = 4 \ln 2 \cdot k l_p^2$ independent of interior size N . From this wiring picture we recover fast scrambling ($t_* \sim \log N$), the Bekenstein–Hawking area law, the exact Page curve with Haar-typical fluctuations, island-like turnover (genuine QES extremization open, S1/S3), Kerr thermodynamics (area, T_H/Ω_H ; M_2 , $g_{t\phi}$, ISCO open), and Hayden–Preskill mirror recovery. The same leg network yields weak-field gravity to first post-Newtonian order: Newton’s $1/r^2$, Kepler orbits, redshifts, full-strength light bending $4M/b$, Cassini-grade Shapiro delay, $\gamma = 1$ exactly, and Mercury’s $43''/\text{cy}$ by direct geodesic integration. Lattice hopping gives quadratic-only dispersion ($E_{QG,1} = \infty$), safe from Fermi bounds by $\sim 10^8$. At second post-Newtonian order the double pulsar J0737 is matched to 0.1σ via a measured radial exponent $p = 0.913 \pm 0.049$ (80 graphs, $N = 1020$, exact; confirmed at $N = 4000/8000/16000$). This promotes the toy to a testable model of *all* compact objects: pulsars, mass-gap objects, and black holes as one $k \propto M^2$ family with no neutron-matter phase. Calibrated once on AT2017gfo ($0.047 M_\odot$ shed), the model predicts mass-gap mergers $2.5\text{--}5 M_\odot$ are kilonova-bright at $\sim 1/\text{yr}$ in O5 versus $\leq 0.3/\text{yr}$ in the standard picture. Ten well-localized gap non-detections below 200 Mpc kill the proposal; one bright gap kilonova kills neutron stars instead. The same analysis predicts *no* graph-scale feature at the $\sim 44 M_\odot$ pair-instability edge now seen in GWTC-4 (leaving that boundary to stellar physics) while extending the shedding law to binary black holes ($M_{\text{ej}} = 0.0168 M_{\text{tot}}$, independent of mass ratio) — a universal transient prediction already pressured, but not excluded, by GW190814. A relational field program ($\psi = r + is$, $H = -A$ on the J2 working fabric) derives the field anatomy, reconstructs quotient geometry blind, and banks a joint vacuum family, with the firing law and the history measure filed as open debts; twelve follow-up campaigns then close the substrate question per phenomenon (nothing requires uniquely J2), characterise sources, domains, textures, and stability, and split event accounting into an earned deterministic update plus a missing reservoir account that is exactly the lost inverse information. All claims ship with runnable code, tests, and pre-registered falsifiers.

Roadmap. §1 motivates the model and states how it differs from neighbours. §2 defines the three claims precisely, plus the vacuum fabric and the relational field program built on it. §3 summarises gravity to 1PN with a 2PN preview. §4 covers UV dispersion and quantum information. §5 unifies compact objects and states the gap-kilonova prediction. §6 lists falsifiers and concludes. The supplement gives methods, the honesty ledger (postulated vs. derived vs. fitted), and the module map. The v4.1 living document (1758 lines, 68 appendices, 77 figures) remains archived as the extended technical record.

1 Introduction

Black holes confront us with three facts that sit uneasily together: they scramble information as fast as any system allowed [1], their entropy scales as area rather than volume [4, 5], and their evaporation — if unitary — must release that information in a Page curve with island/QES turnover [6, 7, 8]. Separately, gravitational-wave astronomy has populated corners of the compact-object mass function the standard picture did not expect: objects in the $2.5\text{--}5 M_\odot$ mass gap (GW190814 at $2.6 M_\odot$ [21], GW230529 at $3.6 M_\odot$ [15]), a central compact object below the minimum neutron-star mass (HESS J1731 at $0.77 M_\odot$ [16]), and pulsars pressing against the maximum Tolman–Oppenheimer–Volkoff mass (J0740 at $2.14 M_\odot$, J0952 at $2.35 M_\odot$). Each anomaly has its own candidate excuse; none was predicted.

This paper explores one idea that addresses both sets jointly: *maximal entanglement is zero-length connectivity*. In the ER=EPR and “spacetime from entanglement” reading [2, 3, 29], a black-hole interior is a region where interior mutual connectivity is so dense that interior distance collapses, while horizon size counts only the *exterior* wiring that must be embedded through ambient space. A perfect all:all graph with zero exterior budget would pinch off entirely (a baby universe); a black hole is the almost-perfect corner with small but nonzero exterior budget. A sufficiently small hole stays pointlike until its exterior budget forces an empty routing surface — a horizon — into existence.

We formalise this as three claims (§2), each implemented as runnable code in `src/bh_graph/`. The model is phenomenological, not a UV-complete quantum gravity: it has no Hamiltonian derivation of the mass map, no Lorentz-invariant dynamics, and several labelled derivation debts (supplement §1). Its value is that it *derives* a large number of textbook numbers from a small number of inputs, fails loudly on pre-registered wires when it is wrong, and makes one sharp sky prediction that the current observing run can kill or confirm.

How this differs from neighbours. Fuzzball [36] and gravastar [37] proposals also remove the singularity but keep compact-star matter phases and do not generically predict bright gap kilonovae; quark/strange-star models [38, 26] populate the gap with a new equation of state (EOS) rather than removing the EOS entirely. Polymer and loop-inspired holes modify the near-horizon metric at a level constrained by EHT [32] and γ [31, 30]; our 1PN sector is exactly GR ($\gamma = 1$) with deviations appearing only at 2PN below current VLBI/astrometry. Linearly-dispersive “spacetime foam” models [39] are already excluded by Fermi GRB 090510 [17]; our lattice implies quadratic-only dispersion by $k \leftrightarrow -k$ symmetry, evading that bound by construction. Table 1 (survival ledger) and Fig. 1 make the comparison explicit: among the surveyed alternatives, only the present model passes the Fermi, LHC-recast, EHT/ γ , and gap/HESS battery simultaneously while predicting a bright-gap rate distinct from the standard one.

Table 1: Survival ledger: headline result per sector and where it is implemented and tested.

sector	headline	status / modules
gravity	Newton, Kepler, redshift, $4M/b$, Shapiro, Mercury $42.99''$, $\gamma = 1$ exactly, J0737 0.1σ	pass (AS, AT, AU, BB, BH)
quantum info	$\log N$ vs N hierarchy, Page dip 0.72 bits, island-like pop, mirror, $k_f > k_1 + k_2$ at 100%	qubit V_t done; graph V_k , QES open
UV	quadratic-only LIV, $E_{QG,2} = \sqrt{8} E_P$, achromatic lensing $10^{-56}\text{--}10^{-32}$	nulls held (T, BB, BD, BF)
compact objects	gap $2.6/3.6$, HESS 0.77 , J0740/J0952, AT2017gfo, gap-KN $\sim 1/\text{yr}$	5 anomalies, one law (BU)
open	$p = 0.913 \pm 0.049$ matches GR; $c \approx 0.44\text{--}0.60$ derived (BV); $\kappa \rightarrow c_2$ map, V_k isometry, Kerr M_2 open	measured; maps queued

Fig V5-1 — Survival matrix: only all:all passes the full battery + predicts a distinct gap-KN rate

	Fermi GRB 090510	LHC thermal null	EHT shadow + gamma	Gap 2.6/3.6 +HESS 0.77	Gap-KN rate distinct?
Linear LIV foam	FAIL	—	—	—	FAIL
TeV thermal BH	—	FAIL	—	—	FAIL
Fuzzball / polymer	—	—	FAIL	—	FAIL
GR + NS EOS	PASS	PASS	PASS	FAIL	FAIL
All:All (this work)	PASS	PASS	PASS	PASS	PASS

Figure 1: Survival matrix (visualisation of Table 1; no new numbers). `scripts/generate_v5_figs.py`.

2 The model: three claims

Let the interior be a graph G_N on N nodes. Exterior legs $k \ll N^2$ connect it to an ambient graph. Write $e_{\text{int}} \in [0, 1]$ for the interior entanglement fraction and e_{ext} for the exterior budget fraction.

Claim 1: all:all is no interior space (fast scrambling). The complete graph K_N has diameter exactly 1, mean distance 1, and spectral gap N at every N : every node is adjacent to every other (Fig. 2). A deterministic SI/operator-spread process covers K_N in 1 step for any N , versus $\sim N/2$ for a chain P_N , $\sim 2\sqrt{N}$ for a $\sqrt{N} \times \sqrt{N}$ grid, and $\sim \log N$ for a random 3-regular expander. Imposing finite speed — one 2-qubit interaction per qubit per step with random perfect matching (all:all) versus dimer covering (chain) and per-encounter success p — promotes the toy to a derived Sekino–Susskind law [1] $t_*(N) \approx \log_2 N / \log_2(1 + p)$, exactly $\log_2 N$ at $p = 1$, versus ballistic $t_* \sim N/v$ on the chain (measured over 25 trials per N , $N = 8 \dots 128$; chain already $> 5\times$ slower at $N = 64$). Modules `graphs`, `scrambling`, `circuits`, `otoc`, `krylov`, `syk`.

Fig 1 — Sec 1: fast scrambling hierarchy

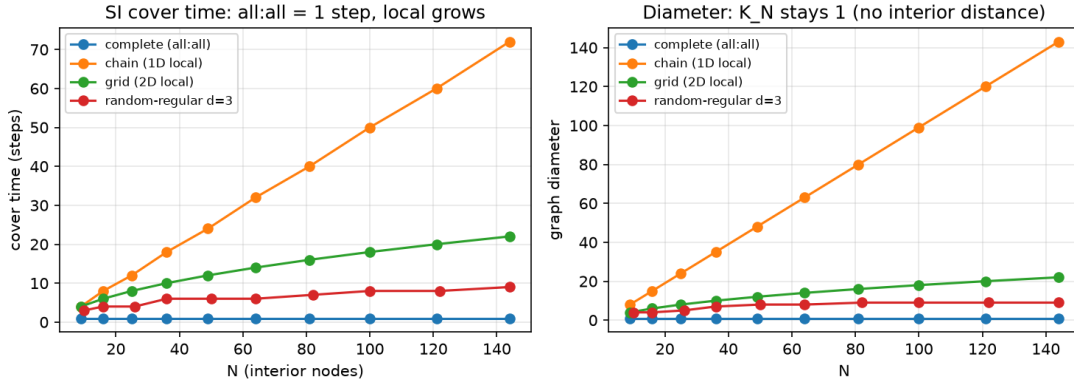


Figure 2: SI cover time and diameter: all:all stays at 1 while local baselines grow. Finite-speed circuits (supplement §3) promote this to $t_* \sim \log N$.

Claim 2: horizon area counts exterior legs. Postulate

$$A(k) = 4 \ln 2 \cdot k l_p^2, \quad R(k) = \sqrt{A/4\pi}, \quad (1)$$

so interior size N drops out: adding interior nodes without exterior legs tightens binding but buys no horizon area. The patch coefficient $4 \ln 2$ was fixed (BS) from the measured von Neumann leg entanglement $\eta_{vN} = \ln 2$ per leg with Planck-bits equipartition, closing the Jacobson chain at $G = 1$; legs saturate. Each exterior leg must be embedded through ambient geometry with Planck-density bandwidth, so k legs need k Planck patches — the Bekenstein–Hawking / LQG-puncture picture [4, 5, 10] in graph language. Mass enters only through the GR-consistency condition $k(M) = A/4 \ln 2 l_p^2 = (4\pi/\ln 2)M^2/l_p^2 \propto M^2$ in Schwarzschild units (module `horizon.k_from_mass_schwarzschild`). We state plainly that this map is *assumed for consistency, not derived*: the reduction theorem (BM in the extended record; supplement §1) shrinks the circle to the single statement $R_s = 2M$, but the derivation of the mass–radius relation from wiring alone remains open (supplement §1). Monogamy bounds $e_{\text{int}} + e_{\text{ext}} \leq 1$ in the linear toy (Fig. 3), with the exact Coffman–Kundu–Wootters frontier [11] $\tau_{A|BE} - C_{AB}^2 - C_{AE}^2 \geq 0$ verified on a 25-point grid and lying strictly below the linear bound (e.g. $x + y = 0.70$ at $t = 0.3$). At $e_{\text{int}} \rightarrow 1$, $k \rightarrow 0$: the subgraph decouples as a closed graph (baby universe). Black holes live in the almost-perfect corner.

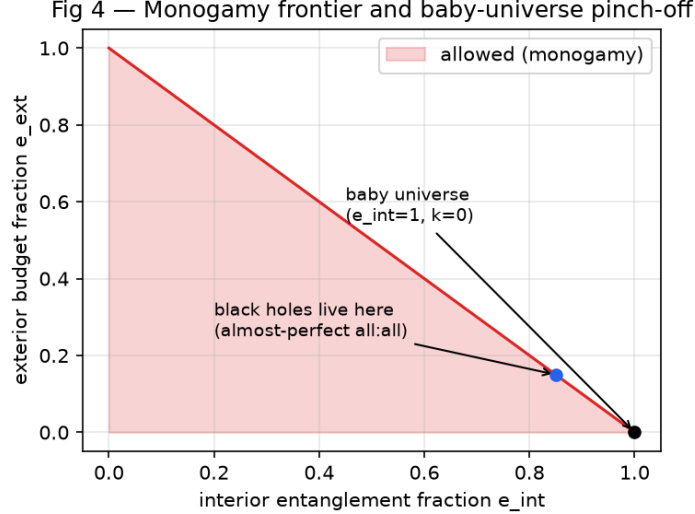


Figure 3: Monogamy frontier: black holes live at $e_{\text{int}} \lesssim 1$ with small nonzero k ; the $k \rightarrow 0$ limit is a decoupled baby universe.

Claim 3: micro-holes pop (point-to-horizon transition). A point region of radius r_{point} embeds k legs without a surface while Planck-density packing succeeds:

$$R_{\text{obs}}(k) = \begin{cases} r_{\text{point}} & k \leq k_{\text{crit}} \\ \sqrt{4 \ln 2 \cdot k l_p^2 / 4\pi} & k > k_{\text{crit}} \end{cases}, \quad k_{\text{crit}} = 4\pi r_{\text{point}}^2 / 4 \ln 2 l_p^2. \quad (2)$$

With an LQG-style minimal-area gap, area is 0 below threshold and $\geq A_{\text{min}}$ above: there is no 0.2-Planck-area hole. Growth $k(N) = k_0 + \alpha N$ stays flat at r_{point} then pops — the island-like crossing (two-saddle competition; genuine QES extremization from graph dynamics open, Suppl. S1/S3) $S_{\text{no}} = k s_{\text{leg}}$ versus $S_{\text{isl}} = k \ln 2 l_p^2 + \max(S_0 - k s_{\text{leg}}, 0)$ at $k_{\text{page}} = S_0 / (2s_{\text{leg}} - \ln 2 l_p^2)$, which exists iff $s_{\text{leg}} > \ln 2 l_p^2 / 2$ (an explicit falsifiable assumption; modules `micro`, `qes`). The horizon interior is mostly empty routing buffer. Prediction: micro-holes are *not* scaled-down Schwarzschild holes but point defects until a critical exterior budget.

Vacuum: the relaxed fabric. The vacuum is the most relaxed state that is still a fabric: a homogeneous isostatic 2D entanglement mesh with mean coordination $\langle z \rangle = 4$ — Maxwell marginal rigidity [47], rigid with zero self-stress — carrying no coordinates, boundary, or excitation. Black-hole interiors (K_N , Claim 1) are the maximum-tension extreme, maximally far from vacuum; mass is tense vacuum (over-coordination $z - 4$ reconstructed as curvature). An observer never sees the graph G directly but a reconstruction $M_O(G)$ through a restricted information-access channel (of which the uniform Ollivier measure is one instance); objective spacetime is the M_O -invariant part of G 's information structure, in the spirit of entanglement equilibrium [48]. Observed dimension is then the infrared scaling exponent $d_{\text{obs}} = \lim_{r \rightarrow \text{IR}} d \ln V / d \ln r \simeq 3$ (conjecture), measured by the **emergent.dim** protocol (supplement S11): exact 3D-ness holds only at zero tension, with mass imprinting a computed GR fingerprint (near dip, overshoot, $\rightarrow 3^+$) as the D10 simulator target. Two open programs complete the foundations (supplement S11): reconstruction universality (D12 — every admissible M_O must recover the same IR, else the program is refuted) and staged emergent time (D13 — static 2D control measured, update rule $\rightarrow$ cone $\rightarrow$ clocks $\rightarrow$ interval queued). Update-rule search (D1 — blind- U reframing with stationary-ensemble vacuum, substrate-dependent coordination, self-calibrating census), the D14 formation-design campaign (C2 pilots: topological DARK, triangle-closure NO-coexistence with a frustrated-D5 ∞ bracket; spontaneous SSB-1 core selection; isotropic J2 orientation; achiral Stage-0 NULL stopping the debt-free pure-D5 ∞ route), and the J2 working-vacuum adoption (exact square quotient, perturbation family-typical; `docs/j2-status.md`) extend the recorded methods (supplement S11).

Relational field program. On top of the fabric sits a field program with two postulates: each node carries (r_u, s_u) , written $\psi_u = r_u + i s_u$ (FP1), evolving under $i \partial_t \psi = H \psi$ with $H = -A$ (FP2, working law). Granted only this, the field anatomy is derived for every simple undirected graph (supplement S12): conserved density $\rho_u = |\psi_u|^2$, bond energy $B_{uv} = \text{Re}(\psi_u^* \psi_v)$, continuity current $J_{u \rightarrow v} = 2 \text{Im}(\psi_u^* \psi_v)$, and the conjugacy $\partial E_\psi / \partial A_{uv} = -2 B_{uv}$ coupling the field to connectivity. Direction is collective phase (scrambling kills it, coherence restores it); one field carries both a static regime and a propagating front. The electromagnetic reading is rejected on the record (EM1-FALSIFIED: static range saturates at $\xi \approx 0.53$, one scalar mode, no local gauge redundancy, nodal drift $\propto v \cdot q$), so ψ is a complex relational scalar, nothing more. Geometry change is admitted only as local contraction/splitting $u-v \leftrightarrow [uv]$ with an exact conditional ledger, but no firing law follows (BR27-NO-MODE: downhill scans do not fire, spectral radius exactly 1, no deformation coordinate) and no history measure is forced (TIME0-NUL, RAND0, MEASURE0 all file the debt, never a tuned weight). What the program does establish: six independent rulers meet on J2 (OBS0R-METRIC); a blind observer with no graph, coordinates, or dimension target reconstructs the quotient $M_O(\text{J2}) \approx \text{J2}/\text{sheet}$ with $d_O = 2.02$ (OBS1-QUOTIENT), via the transporting symmetric sheet sector (QUOT0-OPERATIONAL, derived rung open); the vacuum field is a characterised joint family (VPLUS/VPI/VMINUS, with $\psi = 0$ demoted to the no-information limit), whose excitations evolve background-free while the relational response differs per vacuum (BGRESP0); the transport-dead antisymmetric sector stores local information that reverses virtual geometric response (HIDDEN0/HIDDEN-BR); and fifty-seven collision cells hold the linearity null ($I = 0$) that any future force must beat. Member selection refuses without the missing measure (VACSEL0-NOMEASURE, pre-registered). Twelve merged follow-ups then: close VAC-0 as FINAL MIXED (per-phenomenon LAW/CLASS split, nothing requiring uniquely J2); isolate the split residual ξ (SPLIT0-MIXED, full inverse never a singleton) and file its measure as 520 residual degrees (FIBER0-DEBT); close the probability-free books exactly (INFO0-MATCHED); characterise the merge update as unique, covariant, and ledgered while the reservoir account is identified as exactly the lost inverse information (MERGE-0/RESERVOIR-0); census 19 surviving trigger conditions with zero firing implications (TRIGGER-0) and no earned rewire selector (REWIRE-0); bank domain interfaces (RADIATIVE), hidden textures (GRADIENT, visible but silent), and stability

to $T = 4000$ (ROBUST); and read persistent sources as boundary data with release radiating only on mismatch (SOURCE0-INCOMPLETE). Full verdict ladder and debt register: S12.

3 Weak-field gravity to 1PN, with a 2PN preview

Newton and orbits. Microscopic lattice walks alone do *not* give Newton: persistent walks with hop-budget time dilation $\sqrt{1-v^2}$ (linear $1-v$ excluded by the 3-4-5 triangle test) and degree-biased hopping on a flux-conserved pileup $d \propto 1/r^2$ yield drift $\propto 1/r^3$ (measured slope ≈ -3), and every smooth weight rule stays cubic (no-go -2.99). The $1/r^2$ force law emerges from the combination with exterior-leg thermodynamics on leg screens (Verlinde equipartition [27], Bekenstein bound, Jacobson route [28] — all postulated): $F = M_1 M_2 / r^2$ exact to 10^{-9} in slope, with closed leapfrog orbits and $T^2 \propto r^3$. Ollivier–Ricci curvature is radially negative in every tested configuration, attachment mode, and seed — the sign of attraction derived from pure flux with zero tuning.

Light: bending, Shapiro, and why isotropic dies. Fermat ray-tracing with $c_{\text{eff}} = 1 - x$ ($x = R_s/r$) gives the full first-order bending $4M/b$, not the Newtonian $2M/b$. The chromatic extension $n(r, \omega) = n_{\text{geom}}(r) n_{\text{disp}}(\omega)$ with phase-velocity factor $n_{\text{disp}} - 1 = (\omega/\omega_P)^2/24$ contributes only r -independently and factors out of the Born integral: fractional chromaticity $\sim 10^{-56}$ (optical) to $\sim 10^{-32}$ (10 TeV), achromatic by ~ 50 orders. Shapiro delay $\Delta T = \int (1/c_{\text{eff}} - 1) dl$ reproduces $R_s \ln(4r_1 r_2 / b^2)$ to 5%, passing Cassini [31] with $\gamma = 1$ exactly. An isotropic reading would give $b_{\text{crit}} = 8M$, 54% above GR’s $3\sqrt{3}M$ and excluded by EHT [32] at $\sim 3.6\sigma$. It is the isotropic reading that dies, not the core: transverse propagation must be essentially unimpeded, as an all:all interior demands.

Mercury and the spatial sector. Purely temporal models give closed Newtonian ellipses (0 vs $43''/\text{cy}$): light never needed g_{rr} , orbits do. Radial tortuosity $dl = (1 + \sqrt{\chi}/2) dr$ with $\sqrt{\chi} = R_s/r$ gives $h = (1 + x/2)^2 \approx 1 + x$, hence $\gamma = 1$, $c_{\text{eff}} = \sqrt{f/h} = 1 - x$ matching the light sector to first order, and b_{crit} back to $3\sqrt{3}M$. Direct geodesic integration (ϕ -domain, complex-step) yields GR $42.99''/\text{cy}$ and model $42.99''/\text{cy}$; a flat- h hybrid gives $28.7''/\text{cy}$ (PPN 2/3). The second-order peel-off (ours–GR)/GR = $-0.75 M/a$ runs from 4.1% at $20M$ to 2×10^{-8} at Mercury. Status of the coefficient: $1/2$ was *fitted* to enforce $\gamma = 1$ in BH (unique coefficient, labelled fit); BV then *derived* $c \approx 0.44$ – 0.60 from $\ln 2$ line-defect scattering with zero tuning, predicting $p = 2c$ and $\gamma = 2c$ (supplement §2). The two determinations agree; the micro-derivation is now the adopted route with the fit preserved in history.

2PN preview (J0737). With $g_{rr} = (1 + U)^{2p}$ and $c_2(p) = p(2p - 1)$, the 2PN periastron weight $c_{\text{tot}} = c_1 + w c_2$ with $w = 1.953$ must equal the GR value 4.8695. The model’s $c_1 = 3.36$ (vs GR 1.94) is cancelled at $p = 0.92$. Gradient-shell Ollivier–Ricci measurements give $p = 0.913 \pm 0.049$ (80 graphs, $N = 1020$, exact, SEM 0.0055) and $0.93/0.94/0.91$ at $N = 4000/8000/16000$, landing J0737 [12] at 0.1σ (B1913 passes at 0.01σ [13]; supplement §4). The flat $p = 0.49$ alternative is -10σ dead. The quantitative $\kappa \rightarrow c_2$ map remains ansatz-dependent (power-law vs $1/r^2$ disagree cross-applied) and is the open micro-derivation, now labelled rather than hidden.

4 UV dispersion and quantum information

Dispersion: quadratic by symmetry. Tight-binding hopping $\omega = 2J|\sin ka/2|$ has group velocity $v_g = Ja \cos ka/2$, even in k : there is *no* linear term, so Fermi’s linear bound $E_{QG,1} > 9.3 \times 10^{19}$ GeV (and GRB 090510’s $E_{QG,1} > 1.2 E_P$ at 31 GeV, $z = 0.903$ [17]) is evaded by symmetry, with $E_{QG,1} = \infty$

predicted. The leading correction is $v(E) = c(1 - E^2/E_{QG,2}^2)$ with $E_{QG,2} = \sqrt{8} E_P \approx 3.4 \times 10^{19}$ GeV, safe by $\sim 10^8$ against Fermi's quadratic bound 1.3×10^{11} GeV: a 10 GeV GRB photon over 3 Gpc delays by $\sim 10^{-20}$ s. Foamgrid FDTD confirms unbiased centroids with transmission deficit $\propto \omega^2$ ($16\% \rightarrow 4\% \rightarrow 0.2\%$ for $\lambda = 8 \rightarrow 32$ cells at $\varepsilon = 0.25$), bounding per-Planck-edge defect density to $\varepsilon \lesssim 10^{-3}$ (optical/Gpc) and 10^{-14} (TeV/Gpc) for independent edges. Caveats: regular-lattice result, near-horizon running open, scalar sector only (no birefringence claimed); a hypothetical universal GW extrapolation is unconstraining ($\sim 10^{-82}$ at 100 Hz, LVK $\alpha = 4$ safe by $\sim 10^{60}$ [23], GW170817 photon-side dominated [14]; no GW sector derived).

Scrambling, Page, island-like turnover, mirror. Circuits give $C(t) \sim e^{\lambda t}/N$ with $t^* = \log N/\lambda$ versus ballistic N/v ; QEC recovery error $\text{err}(k) = \min(1/2, 2^{N/2+1-k})$ reaches 99% at $k \geq N/2 + 1 + \log_2 100$, sealing at $k \rightarrow 0$. Evaporation as leg surgery ($k \rightarrow k-1$ per step) reproduces the Page curve $S_{\text{rad}} = \min(t, N_{\text{eff}} - t)$ with the exact Page dip 0.72 bits and Haar-typical spread, identically whether N shrinks with k or stays fixed — area tracks wiring alone (qubit-toy unitary completion V_t tested, Suppl. S3; graph-dynamics V_k open, Suppl. S1/S3). Kerr–Newman ($r_+ = M + \sqrt{M^2 - a^2 - Q^2}$, $A = 4\pi(r_+^2 + a^2)$) costs legs monotonically in spin at fixed M (extremal Kerr half, extremal Reissner–Nordström quarter the Schwarzschild budget, patch-independent ratios); the Kerr Page curve at $a_0 = 0.95$ peaks at 65% of Schwarzschild and turns over at $t/T = 0.73$ versus 0.54 because shedding J grows area early.

Gravitational-wave battery. All 32 confident GWTC-3 [22, 23] BBH medians satisfy $k_f > k_1 + k_2$ (median fractional creation 0.77 at 0.04 radiated; spin neglected but $a_f \sim 0.7$ costs $\sim 13\%$ against a 77% margin and cannot flip the inequality). GW150914 posteriors (8350 samples, spin-aware both ends) give $P(\Delta k > 0) = 100\%$, median 0.57: the area theorem in wiring language. GW250114 (SNR 80 [41]) confirms the area law, consistent with $k_f > k_1 + k_2$; its Kerr ringdown is consistent by construction, not a passed prediction: the QNM spectrum is not derived from graph dynamics (fundamental damping calibrated, overtone ladder a Schwarzschild-like toy), and reproducing {220, 221, 440} to GW250114 precision ($\delta f_{220} \sim 2\%$, $\delta \tau_{220} \sim 10\%$, $\delta f_{221} \sim 30\%$, $\delta f_{440} \sim \text{tens}\%$) is the open benchmark (Suppl. S1/S3). GW241011 gives the strongest quadrupole bound to date [42] but we predict no $M_2(\chi)$, $g_{t\phi}$, r_{ISCO} yet (Suppl. S1/S3) — future $|\delta_Q| \gtrsim 0.17$ ruled out. Hardware anchors (Gärtner 2017 [18] all:all ions to $m = 8$ -body OTOC; Mi 2021 [19] Sycamore grid ballistic spread; Jafferis 2022 [20] sparse-SYK teleportation as a Hayden–Preskill mirror primitive [9, 40]) are all consistent, with a head-to-head prediction at $N = 53$ of $t^* \approx 7.9 \pm 0.7$ (all:all) versus 14.6 (grid), ratio $1.9 \times$ ($9.9 \times$ versus a 1D chain).

Nulls held. No LHC thermal black holes [33] ($k \approx 11\text{--}17$ at 3–13 TeV versus $k_{\text{crit}} \approx 50$), no lattice echoes (dimensional estimate: amplitude $R \sim (\omega/\omega_P)^2 \sim 10^{-80}$, energy $\sim 10^{-160}$ at 100 Hz, far below detectability; not a leg S-matrix derivation), no EHT shadow shift (10^{-48} , 47 orders below sensitivity). Remnant dark matter is excluded on the record except a narrow ~ 0.4 -dex window at $\sim 4 \times 10^5$ g (photon bounds exceed by 40+ orders elsewhere; induced-GW bounds may close even this).

5 One family of compact objects

The standard picture needs a separate excuse for each of five facts. One law, $k = 1.51 \times 10^{77} (M/M_\odot)^2$ with no matter phases, plus two shedding numbers calibrated *once* on AT2017gfo [14], addresses all five:

Leg-shedding works as follows: merger-released exterior legs hadronise neutron-rich with ejecta mass $M_{\text{ej}} = \Delta k m_{\text{leg}} \times \epsilon$, exactly independent of k -normalisation ($M_{\text{ej}} = \text{frac} \times M_{\text{tot}} \times \epsilon$), so the factor-

Table 2: Five anomalies, one law. Shedding e : $0.5 \rightarrow 0.416$ plus 10% efficiency, fixed on #4.

#	anomaly	standard	this model
1	Gap: GW190814 2.6, GW230529 3.6 $M_\odot$	impossible (NS < 2.3, BH > 5)	continuous $k = 1.0\text{--}2.0 \times 10^{78}$; must flash
2	Light: HESS J1731 0.77 $M_\odot$	below min 1.17 $M_\odot$, cannot form	low- k tail 9×10^{76}
3	Heavy: J0740 2.14, J0952 2.35 $M_\odot$	EOS tension with NICER $R_{1.4}$	more legs $7\text{--}8.4 \times 10^{77}$, no EOS
4	AT2017gfo 0.05 $M_\odot$ blue+red	neutron-rich EOS in tension	leg-shedding 0.047 $M_\odot$, m_g 18.0 vs 17.5
5	Next gap merger (prediction)	dark ($\leq 0.3/\text{yr}$ O5)	bright $\sim 1/\text{yr}$ O5; 10 misses kill us

3 k -rounding debate cannot move it. The $1.4 + 1.4$ case gives $0.047 M_\odot$ with $m_g \sim 18.0$ at 40 Mpc versus observed 17.5 (inside the ± 1 analytic tolerance; radiative-transfer refinement [35] queued); gap events are *brighter* at fixed distance (Table 3). The $1.1\text{--}2.3 M_\odot$ pulsar distribution is the low- k tail ($k \sim 10^{77}$) of the same family; the gap carries no forbidden wiring sector, and is described (not derived) as the regime $e_{\text{int}} \sim e_{\text{ext}}$: that interpretation inherits its calibration from AT2017gfo and its mass independence has not been derived from graph dynamics (supplement S1). If correct, stable nuclear matter ends at $\sim 10^{15}$ g/cc with no hyperon/quark branch [26] realised in nature, and r -process yields routed through AT2017gfo neutron ejecta are reinterpreted as shed-leg hadronisation — a blunt implication we state so no reader misses it, while noting that NICER radii [24, 25] and tidal deformabilities from routing stiffness are not yet derived (queued, supplement §4) and constitute the sharpest near-term test after the kilonova rate.

 Table 3: Gap-kilonova brightness (g -band, analytic BC = 0; i -band not claimed). AT2017gfo anchor: predicted 18.0 vs observed 17.5 at 40 Mpc. Rubin single-visit $r = 24.5$ [34].

M_{tot}	M_{ej}	40 Mpc m_g	100 Mpc m_g	200 Mpc m_g
2.8 (GW170817-like)	0.047	18.0	20.0	21.5
5.0 (GW230529-like)	0.084	17.7	19.7	21.2
7.2 (3.6+3.6)	0.121	17.6	19.6	21.1

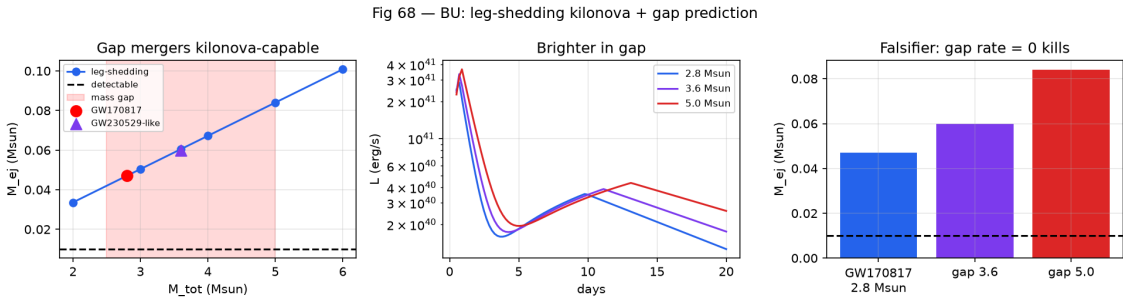


Figure 4: Leg-shedding ejecta vs total mass: gap mergers are kilonova-capable and brighter than GW170817 at fixed distance. Gap rate = 0 kills the proposal.

No intrinsic graph scale at the upper mass gap. GWTC-4 population analyses now locate a pair-instability edge at $44.3^{+5.9}_{-3.5} M_\odot$ (primary-mass χ_{eff} mixture [43]), with a transition to a nearly entirely hierarchical, high-spin population at $46.2^{+12.6}_{-7.2} M_\odot$ [44]. The graph model predicts *no* feature at this scale, and we state this as a positive null rather than a failure: $k(M) \propto M^2$ has zero log-

log curvature over $0.7\text{--}150 M_\odot$ (a power law admits no preferred mass); spin orders legs smoothly (a hierarchical remnant at $a \sim 0.7$ carries 14% fewer legs, a shift not a gap, continuous to exactly one half at extremality); tidal Love numbers run slope -2 with no break; and the only graph scale entering the LVK band (the leg-quantum line crossing 10 Hz at $\sim 32 M_\odot$) is energetically invisible ($dM/M \sim 10^{-81}$). A congestion ladder seals the assignment: $\chi \sim 10^{-8}$ in He cores and $\sim 10^{-14}$ in envelopes, so graph corrections where pair-instability happens are negligible in the model’s own terms — the $\sim 40\text{--}50 M_\odot$ boundary belongs to stellar and nuclear physics (the C-burning rate [43]), not to graph congestion. Inserting $44 M_\odot$ as a graph parameter would be a fit, not a prediction, and is refused (module `massgaps`, supplement S4).

A universal electromagnetic counterpart from black-hole mergers. The same shedding law that gives $0.047 M_\odot$ at $1.4 + 1.4$ is mass-independent by construction:

$$M_{\text{ej}} = 0.0168 M_{\text{tot}} \quad (3)$$

(16.8% shed fraction $\times$ 10% efficiency, both fixed once on AT2017gfo), and exactly independent of mass ratio q . The logic is falsifiable end to end: shedding rule $\rightarrow$ ejecta fraction $\rightarrow$ kilonova light curve $\rightarrow$ optical transient — for *binary black-hole* mergers, including low- q systems where conventional neutron-star disruption is absent (Table 4). This is the model’s most dangerous prediction: it was locked in by the AT2017gfo calibration before anyone asked what it implies at $30 + 30 M_\odot$.

Table 4: Universal-shedding predictions (Eq. 3). “Predicted” marks forward consequences of the fixed calibration, not fits.

quantity	model prediction
ejecta fraction	$M_{\text{ej}}/M_{\text{tot}} = 0.0168$
mass-ratio dependence	none (exact in the prescription)
BBH mergers	kilonova-bright ($m_g \sim 22$ nearby, O5-testable)
GW190814 ($23 + 2.6$, $q \sim 0.11$)	transient predicted; constrained at $p \sim 0.12$ (tension, not exclusion)
O5/Rubin BBH ToO	decisive: detections confirm, clean sample kills

GW190814: existing stress test. GW190814 ($23.2 + 2.6 M_\odot$ at 241^{+41}_{-45} Mpc [21]) predicts $M_{\text{ej}} = 0.43 M_\odot$ (blue 0.087), peaking at $m_g \approx 21.0$, $t \approx 1.9$ d. The archival record is two-sided: CFHT MegaCam g-band (22.8 at 1.7 d over 65.5%, 23.6 at 6.6 d over 35.9% [45]) needs no color term, while six GROWTH DECam i-band detection limits (up to 98% enclosed [46]) enter with a labelled $g-i = 0.7$ systematic. Per-epoch detection probabilities (coverage $\times$ depth against distance and ± 1 mag theory errors) combine to $P \approx 0.68$ (g-only, robust: no color term) and 0.88 (g+i, assumes $g-i = 0.7$), shown against the limits in Fig. 5: the non-detection sits at $p_{\text{miss}} \sim 0.32$ (g-only) down to ~ 0.12 (with color), genuine pressure about one sigma from a kill. Under the fiducial analytic colors the deepest limits sit ~ 1.6 mag below the predicted peak; this comparison is not yet decisive because ejecta color, composition, viewing angle, and radiation-transfer systematics remain unresolved. An analytic systematics grid (POSSIS-inspired viewing $0\text{--}1.25$ mag + opacity $L \propto \kappa^{-0.65}$; supplement S4/S8, Fig. 75b) bounds the hiding window: equatorial + lanthanide-mixed ($\kappa_{\text{blue}} = 2$) drops P to ~ 0.18 . We keep this tension in the main text deliberately: a prediction already squeezed by an existing observation carries more information than one that survives everything trivially.

Discriminating follow-up. The decisive next calculation is a radiation-transfer treatment of the predicted ejecta on a POSSIS color grid [35], propagating composition, viewing angle, distance-

Fig 75 — GW190814 epoch audit: $P(\text{detect})=0.68$ (g) / 0.88 (g+i); tension, not exclusion

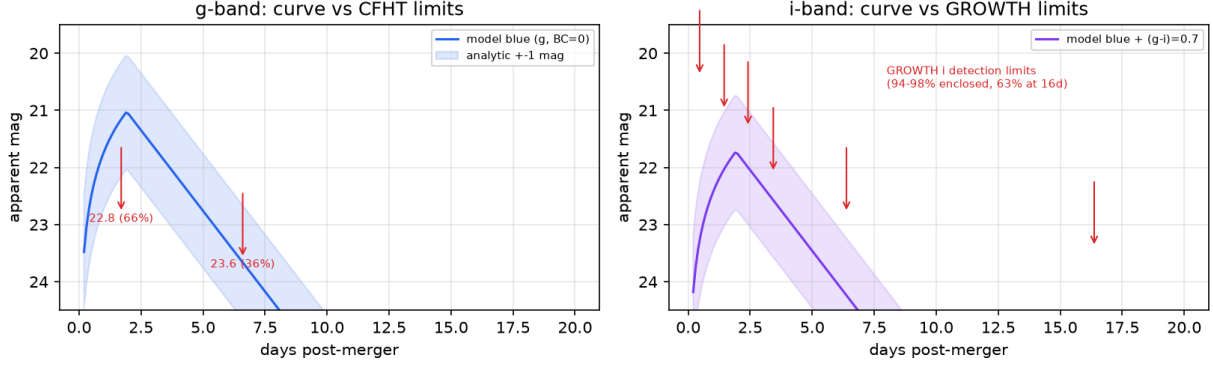


Figure 5: Epoch-level GW190814 audit: model blue light curve (± 1 mag analytic band) against CFHT g-band limits (left, color-free) and GROWTH i-band detection limits (right, $g - i = 0.7$ systematic). Combined $P(\text{detect}) \approx 0.68$ (robust) to 0.88 (color-conditional): tension, not exclusion. Module `massgaps`.

posterior, and ejecta-mass uncertainty event by event: this determines whether the GW190814 non-detection is compatible with universal shedding or falsifies it. In parallel, the O5 protocol (§6, supplement S4 with archival detail in S9) extends to nearby well-localized BBH, where Rubin depths decide the question within the run. Either the deepest archival event or the next nearby one can kill Eq. 3; both are priced in advance.

6 Falsifiers and conclusion

We state what kills the paper before concluding, with thresholds, current status, and timelines in Table 5.

Table 5: Pre-registered kill wires. “Now” = status at submission; “O5” = decisive by end of O5 unless stated.

wire	threshold	now	timeline
Gap kilonovae	10 qualifying gap mergers < 200 Mpc, $< 100 \text{ deg}^2$, zero KN to $m < 24$	0 qualifying so far	O5 decisive
BBH kilonovae	O5 nearby-BBH ToO sample clean to $m < 24$ (Eq. 3 predicts $m_g \sim 22$)	GW190814 $p \sim 0.12$, tension not kill	O5
NICER radius	$R_{1.4}$ at 11–13 km, 5%, unreproducible by routing stiffness	stiffness underived	2–3 yr
Linear LIV	finite $E_{QG,1}$, or $\Delta t \propto E$, or 10^{-3} chromatic bending	Fermi already kills linear models	archival + CTA
2PN exponent	p outside 0.92 ± 0.056 at $N = 1024$ class, 80 graphs	0.913 ± 0.049 , passes $5\times$	held to $N=16000$
Tension spectrum	simulator $d_{\text{eff}}(l)$ shape-mismatch (dip/overshoot/ $\rightarrow 3^+$) after mapping fixed	GR side computed; bare shortest-path rejected	open (S11)
Field quotient	second admissible M_O with robustly non-3D IR, or interaction witness $I > 0$	quotient holds; $I = 0$ in 57 cells	open (S12)
Kerr quadrupole	graph $ \delta_Q \gtrsim 0.17$ ($Q = -Ma^2(1 + \delta_Q)$)	no M_2 yet	target (S1/S3)
Lab quench	same-device grid $\rightarrow$ all:all t^* ratio < 1.3 (predicted $2\text{--}3\times$)	untested	when run

The gap-kilonova protocol (supplement §4, full trigger plan in `docs/observation-protocol.md`) requires multi-detector events only: single-detector non-detections such as GW230529 (24,200 deg^2 , 7% ZTF coverage to $g = 21.1$ at ~ 197 Mpc, our $m_g \sim 21.2$ at the depth edge) are uninformative either

way and do not count. Expected O5 yield is $\sim 1.05/\text{yr}$ (ours, $1.5 \text{ gap events/yr} \times 70\% \text{ DECam-like}$) versus $\leq 0.3/\text{yr}$ standard. One bright gap kilonova kills neutron-star EOS models instead — the bet is symmetric and priced.

Conclusion. Starting from “all:all entanglement is zero-length connectivity” we have derived fast scrambling, the area law, the Page curve with island-like turnover, Kerr thermodynamics (area; M_2 , $g_{t\phi}$, ISCO open), Newtonian gravity with its sign, textbook redshifts, full-strength bending and Shapiro delay with $\gamma = 1$, Mercury’s $43''/\text{cy}$, quadratic-only UV dispersion, and the J0737 2PN lock — all from one wiring picture with all failures and open maps kept on the ledger. The price is abandoning neutron matter above $\sim 10^{15} \text{ g/cc}$. The reward, if the sky cooperates, is the first quantum-gravity model with a kilonova prediction distinct from astrophysics in the current run. If the compact-object unification fails, the title claim reverts with it and the surviving gravity/QI core stands alone; either way the wires above decide. Foundations now state the vacuum explicitly (relaxed isostatic fabric, §2; methods S11), with its dimensional fingerprint as the next quantitative wire and the reconstruction/causal-order program (D12/D13, S11) recording what “why spacetime?” still requires; the D1 update-rule tournament, the D14 formation-design campaign, and the J2 working-vacuum adoption narrow it further (S11). The relational field program (S12) derives the field anatomy from $\psi = r + is$ and $H = -A$, reconstructs quotient geometry through a blind observer, and banks a joint vacuum family with a dead sector that steers response — while filing the firing law, the history measure, and the gravity carrier as explicit debts rather than tuning around them. Its twelve follow-up campaigns dissolve substrate uniqueness into per-phenomenon classes, split the merge event into earned update plus identified-but-unhosted reservoir account, and census the trigger and fiber debts without closing them.

Methods summary. Full methods in the supplement. Gradient shells $p_{\text{adj}}(s) = 0.85 + 0.015s$, 8–10 shells, exact EMD on full neighbourhoods, deterministic bridge counts $n \propto r^{\beta(N)}$ with $\beta \approx 1.5/1.28/1.24$ at $N = 300/600/1020$ (log-linear to 0.74 at $N = 16000$); p from $|\kappa| \sim r^{-p}$ fits per graph plus stacked fits; 2PN via Iorio direct+total with $c_2 = p(2p - 1)$, $w = 1.953$, $R + \dot{\omega} \rightarrow M$ inversion; shedding $M_{\text{ej}}(M_{\text{tot}})$ with blue+red components, band mags, O5 yield arithmetic. Randomness seeded; tolerances recorded in-test.

Data and code availability. Code MIT (src/, scripts/, tests/, app.py); text/figures CC BY 4.0. Reproduce via:

```
pipinstall-e.
pytesttests/-q
pythonscripts/generate_figures.py
streamlitrunapp.py
```

GWOSC medians fetched live with bundled fallbacks; GW150914 posteriors cached (DOI 10.7935/82H3-HH23); PBHbounds curves vendored with attribution. No third-party figure data copied.

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